PROJECT REPORT / NOVEMBER 2026

Bengaluru trade atlas
Ward Wise

A ward-level decision-support tool for investigating changes in Bengaluru's recorded trade-licence activity. Built for the BAICONF Datathon by Team Codyssey, SRM Institute of Science and Technology, Kattankulathur.

345,123
distinct applications

198
historical BBMP wards

20
calendar quarters

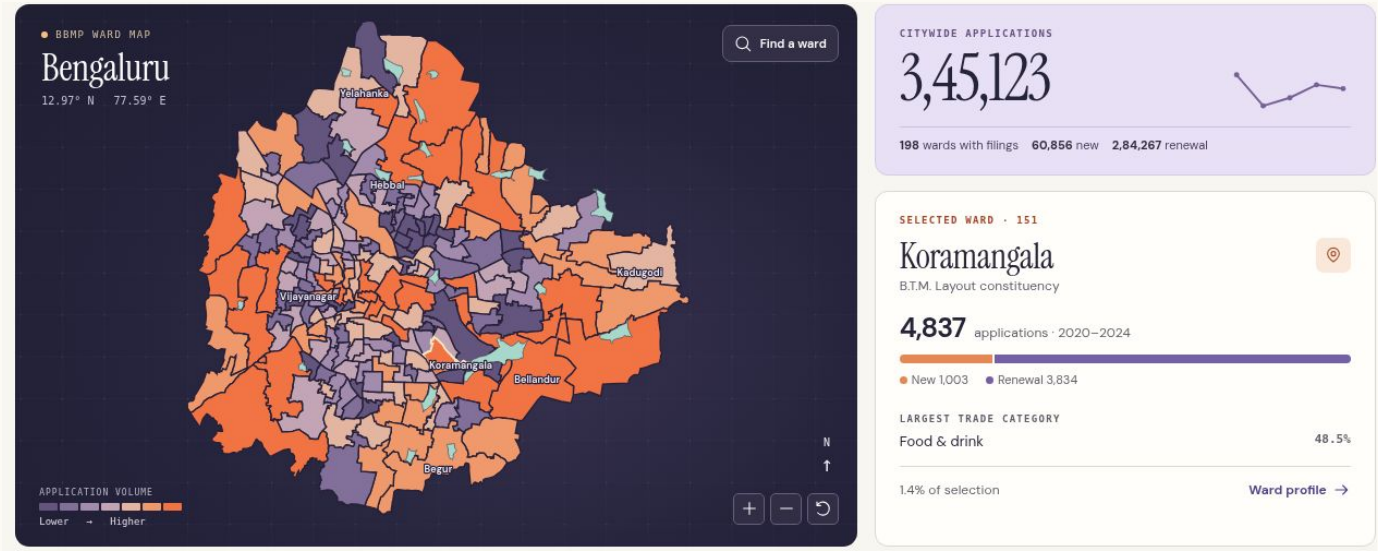


Figure 1. The interactive atlas shows application volume and the selected ward. The five-year view opens with Koramangala selected.

The decision

A business-facilitation officer has limited time for outreach. Ward Wise compares a ward's current mix of applications with its own seasonal history and contemporaneous filings in the other wards. It presents up to three cases per quarter for human review. The site also lets the officer explore every ward, including those without a flagged case.

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DATA AND PREPARATION

From trade lines to filings

The 1 September 2026 public release compiled by Vivek Matthew from the BBMP trade-licence status portal records application type, date, application ID, ward and trade headings. We selected applications dated 1 January 2020 through 31 December 2024. A single application can contain several trade lines.

358,635
source trade line items

13,512
additional lines grouped

345,123
distinct application IDs

Five-year coverage

Year	Applications	Unclassified
2020	77,450	13,817
2021	60,619	9,594
2022	65,028	7,996
2023	72,053	8,271
2024	69,973	6,795

How the records were reconciled

Application IDs agreed on date, type and ward across repeated lines. Each filing contributes to one category. The eight substantive groups follow BBMP’s numbered headings. We retained 5,317 applications spanning multiple headings and 46,473 without a usable heading. The latter comprise 13.47% of the total. Changes in coding can therefore resemble changes in activity.

All 198 source ward names were matched to distinct polygons in DataMeet’s 2012 BBMP boundary file. Of those, 130 matched after simple normalization and 68 required reviewed name aliases. Using one map vintage keeps comparisons consistent across the five years.

The published table contains 19,502 nonempty ward–quarter–category cells, split into 60,856 new and 284,267 renewal applications. Only aggregate counts are served. Ward labels and lake outlines provide orientation. The map’s colour bins reflect relative application volume under the chosen filters.

ARCHITECTURE AND METHODOLOGY

How a case reaches the queue

Stage 1 proposed a digital twin to study ward business mix. The executed pilot keeps Bengaluru, BBMP licence data and the ward as the decision unit. Missing due dates and closure labels led us to use recorded application mix as the measurable outcome. The seasonal statistical detector requires no trained language model, prompt or external inference API.

01 Reconstruct the baseline

Compare the same calendar quarter in earlier years. Estimate the ward’s historical share for the category and shrink it toward the share among the other 197 wards using 40 pseudo-applications. This stabilizes thin series.

02 Adjust for the rest of Bengaluru

Multiply historical local odds by the ratio of the current peer odds to historical peer odds. Convert the resulting odds to a share and multiply by the ward’s total filings this quarter. This estimates the expected count conditional on filing volume.

03 Check evidence and persistence

Require at least 25 historical and current applications, at least five expected inside and outside the category, and two consecutive eligible quarters beyond two working standard deviations in the same direction. The variance uses dispersion 2 and a historical-sample adjustment.

04 Suppress and select

Suppress synchronized ward signals when at least five and at least a quarter of eligible wards qualify for the same trade and direction. Keep up to three remaining cases with the largest absolute standardized gaps and display them alphabetically.

05 Inspect and document

The interface shows the observed count, expectation, model band and peer share. An officer selects one of four interpretations and can add a note. Reviews can be saved in the browser and exported as JSON.

OBSERVED RESULT AND DEMO

A real case in Dodda Nekkundi

In Q4 2024, ward 85 had 142 applications across all trades. Accommodation accounted for 36, or 25.4%. Its historical same-quarter share was 41.76%, while the contemporary share among other wards was 13.41%. The model expected 92.4 after the peer adjustment. The category qualified in two consecutive quarters.

36	92.4	74.5–110.2
observed applications	model expectation	working model band

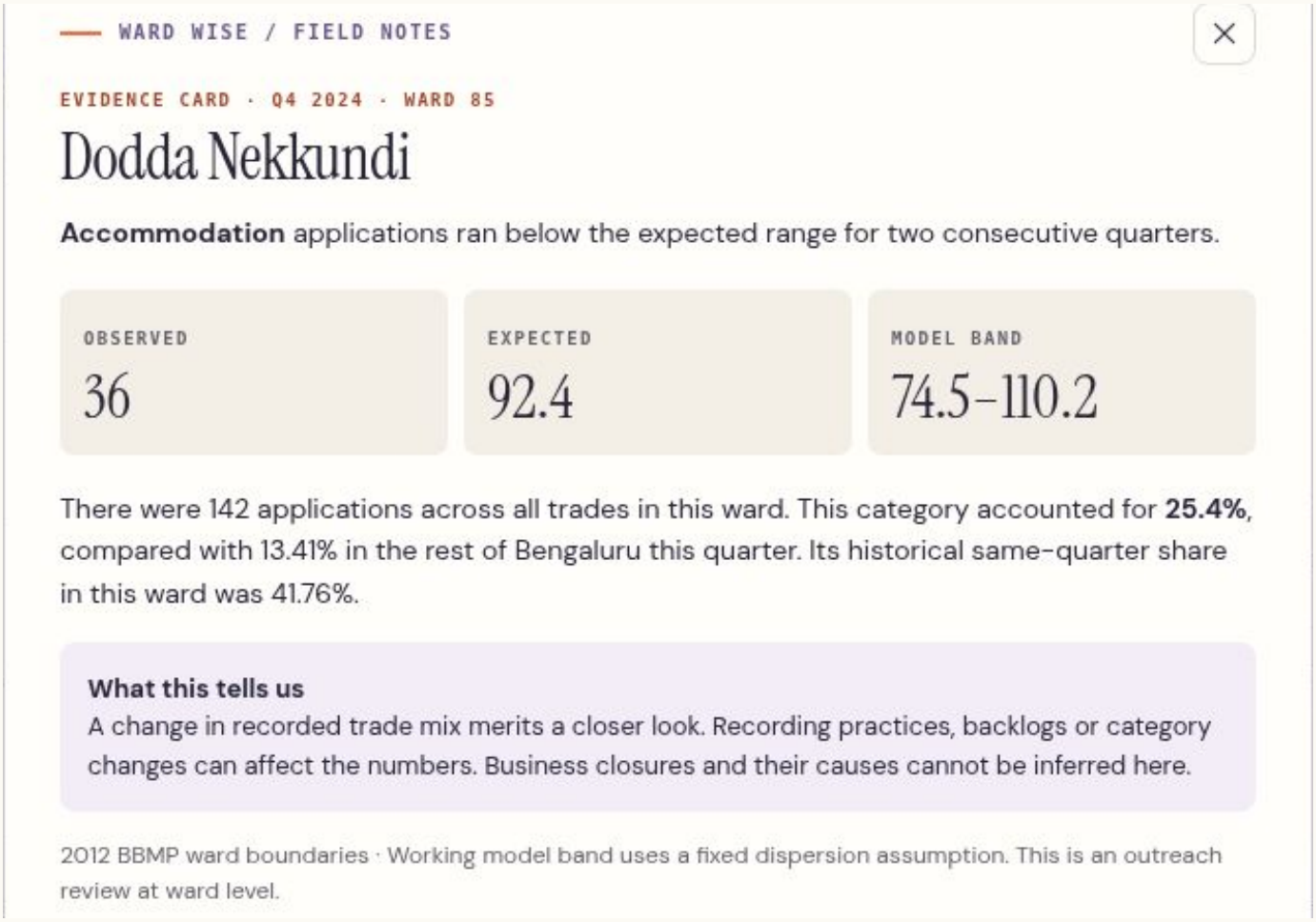


Figure 2. Detail from the live Q4 2024 evidence card. The full screenshot is supplied with the project.

What follow-up could establish

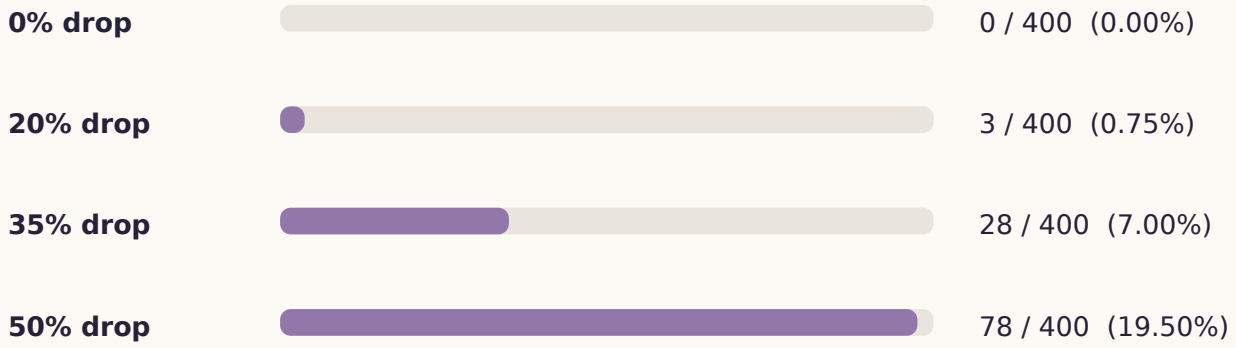
A reviewer could ask whether records were entered late, reclassified or affected by a change in filing practice. A liaison could then seek independent local information. The filing difference alone cannot establish a closure, lost job or decline in local demand.

EVALUATION AND NEGATIVE RESULTS

How much change can we see?

The 2024 historical check uses each quarter’s prior-year baseline and contemporaneous peer counts. To test sensitivity separately, we sampled 400 eligible pairs of consecutive quarters ending in 2024 using a fixed random seed. We simulated counts at the expected share or at a reduced share and applied the two-quarter rule. Each scenario uses the same 400 pairs.

Controlled drops in category share



The 50% drop was detected in 78 of 400 scenarios. A 20% drop was detected only three times. The null scenario produced zero detections in this simulation. These results apply before the top-three cap, under binomial simulation noise. Field false-alarm rates require independent observation.

Persistence compared with a raw count rule

Across Q1–Q3 2024, two of six evaluable selected cases stayed beyond threshold in the same direction one quarter later. A comparator choosing the same number of cases by the largest change in raw filings had zero of six. The horizon excludes Q4, whose next quarter falls outside the dataset. Six cases provide too little evidence for a general performance claim.

Challenges

The extract was assembled retrospectively. Changes in category completeness, backlogs or ward-specific administrative practices can affect the measurements. Working bands assume a fixed dispersion and have no externally measured coverage. A null simulation with zero flags does not show that every real flag reflects economic change.

IMPLEMENTATION, IMPACT AND NEXT STEPS

From the atlas to a pilot

Operation and reproducibility

The client ships static HTML, CSS, JavaScript, fonts, cartographic paths and aggregated data. Python scripts rebuild the public inputs and detector outputs. Seven automated tests reconcile totals, map joins, cell uniqueness, budget limits, a citywide shift and a known evidence card. The repository includes a checksum-locked source, CI checks and Vercel configuration.

Demonstrate the workflow

Open the atlas and hover a ward. Choose 2024, search for Bellanduru and open its profile. The ward records 1,812 applications, including 561 new filings. Then open Review radar for Q4 2024 and inspect Dodda Nekkundi. Save a demonstration review and export it. The screenshot pack includes these views and the methods page.

Economic use and pilot design

With at most three cases per quarter, officers can allocate a small and explicit follow-up budget. Economic impact remains a pilot hypothesis. Record review minutes per case, the distribution of officer assessments and the fraction with independently confirmed administrative or business explanations. Compare time and findings against existing manual review before expanding.

Recommendations

First validate filing and category changes with the relevant civic team. Obtain due-date and validity data before estimating renewal timeliness. Archive dated source snapshots, calibrate working bands on later data and test the detector across wards with sparse categories. A future multi-officer workflow should add authenticated shared review storage.

Sources and licences

BBMP trade-licence archive compiled by Vivek Matthew · ODbL 1.0. Source release dated 1 September 2026. The repository records the source checksum and preprocessing rules.
DataMeet Bangalore municipal spatial data · 2012 ward boundaries, CC BY-SA 2.5 India.
ATREE-CSEI lakes via OpenCity · CC BY. Lake geometry provides geographic context.